

Advanced (Carolina Reaper)

- Q1.** A bookstore has 243 books on a shelf. What is the value of the digit 4 in this number?
- (A) 40
(B) 400
(C) 4,000
(D) 400,000
(E) 4
- Q2.** Round 46,279 to the nearest thousand.
- (A) 46,000
(B) 47,000
(C) 46,279
(D) 40,000
(E) 50,000
- Q3.** A city has 14,753 residents. Write this number in expanded form.
- (A) $1 \cdot 10,000 + 4 \cdot 1,000 + 7 \cdot 100 + 5 \cdot 10 + 3 \cdot 1$
(B) $1 \cdot 1,000 + 4 \cdot 100 + 7 \cdot 10 + 5 \cdot 1 + 3 \cdot 10,000$
(C) $1 \cdot 100 + 4 \cdot 10 + 7 \cdot 1 + 5 \cdot 1,000 + 3 \cdot 10,000$
(D) $10,000 + 4,000 + 700 + 50 + 3$
(E) $10,000 + 4,000 + 700 + 50 + 30$
- Q4.** What is the value of the digit 7 in 94,753?
- (A) 7
(B) 70
(C) 700
(D) 7,000
(E) 70,000
- Q5.** A bakery sells 2,467 cakes per day. Round this number to the nearest hundred.
- (A) 2,500
(B) 2,400
(C) 2,000
(D) 3,000
(E) 2,000
- Q6.** Write the number 94,279 in standard form.
- (A) 94,279
(B) $9.4279 \cdot 10,000$
(C) $9.4279 \cdot 10^4$
(D) $94.279 \cdot 1,000$
(E) $942.79 \cdot 100$
- Q7.** A water tank can hold 150,027 gallons of water. What is the value of the digit 5 in this number?
- (A) 5
(B) 50
(C) 500
(D) 5,000
(E) 50,000
- Q8.** Round 7,463,219 to the nearest million.
- (A) 7,000,000
(B) 8,000,000
(C) 7,500,000
(D) 7,463,219
(E) 8,500,000

Open Ended Questions (FRQ)

- Q9.** A city planner has 85,219 residents to consider for a new development. Write this number in expanded form and round it to the nearest thousand. Show your work and explain your reasoning.
- Q10.** A library has 94,357 books to catalog. Write this number in standard form and then round it to the nearest hundred. Justify your answer with calculations.

Chef's Tips

- Tip 1: Encourage students to use visual aids like place value charts to solve problems.
- Tip 2: Remind students to always read the question carefully and identify the key concepts being tested.
- Tip 3: Emphasize the importance of showing work and explaining reasoning, especially for free-response questions.